Stopping Sets, Saturated Bundles, and Exact Raw Cancellation

A certified performance improvement for the ProveIt unknot-recognition program

Research continuation prepared for Vladimir Reshetnikov
Mathematical development and implementation with ChatGPT

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Abstract

We study two specific bottlenecks in the source-certified normal-cut pipeline. First, we prove that complete positive raw cancellation of an arbitrary rectangular nonnegative incidence matrix is equivalent to deletion of unit column leaves. A nonempty stopping set survives every positive Schur update, which proves completeness of a sparse source algorithm and provides a short obstruction certificate. The canonical peeling construction is classical in coding theory; the persistence argument and its use in geometric cancellation are the contributions developed here. This strengthens the repository's square-only completeness statement, removes repeated matrix scans, and shows that survivor-set enumeration is unnecessary for this decision problem.

Second, we continue the incoming one-two-handle theorem for saturated interval bundles. Their residual attaching curves have a source-computable unsigned incidence on old core cocores, while every added band column is even and can never supply a raw singleton pivot. Thus arbitrarily large numerical base genus can be excluded from cancellation search. We implement a whole-cut certificate that combines the old core rows with one aggregate row per saturated base and independently verifies the resulting handlebody census. All geometric conclusions are tied to authenticated embedded source charts.

The package includes proofs, a repository integration patch with updated files, independent checkers, exhaustive algebraic tests, source-cut comparisons, paired measurements, and a focused research agenda. The improvement concerns supplied-cut certification. It neither implements nor proves a general quasi-polynomial unknot recognizer; surface production, transported boundary patterns, and total hierarchy work remain separate obligations.

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1 Scope, source audit, and the research question

1.1 What this continuation changes

The immediate mathematical question is precise. After choosing a spanning forest in an authenticated geometric handle core, let n_{ih} count the actual intersections of the ith two-handle attaching curve with the cocore of the hth surviving one-handle. The existing implementation accepts a positive singleton pivot $n_{rg} = 1$ and updates

$$n'_{ih} = n_{ih} + n_{ig}n_{rh} \qquad (i \neq r,\ h \neq g). \tag{1}$$

It seeks to remove every two-handle. We ask whether finding a successful order, including its surviving one-handles, actually requires branching.

The answer is no. A unit column leaf is a column whose total incidence on the live rows is exactly one. Its pivot has no fill in the remaining rows. Repeatedly deleting such leaves is complete for the entire positive raw model, even on rectangular matrices. A stopped nonempty row set is a certificate that *no* positive pivot order can finish. This is a stronger statement than a failure of a chosen order.

The second question concerns an interval-bundle component whose entire base boundary is attached to the core. Previous work localizes its obstruction to a single residual two-handle, but the number of added one-handles can be very large. We show that these added columns need never be generated for raw cancellation: they are even. The only relevant row is the aggregate of the source boundary-circle incidences on old core cocores. This connects the exact matrix theorem to a new source-bound geometric operator.

1.2 The pinned repository and incoming reports

The inspected source is the public ProveIt repository, at commit

`704600626606a9097cfd89d4e3d2bbc63c8dd1cb`.

The target subtree is `Topology/UnknotRecognition`; the supplied incoming directory contains four research archives[1, 2]. All source paths and hashes used for integration are retained in the package.

The maintained pipeline already includes compressed normal-cut cells, parallelity bases and their ordered interfaces, source handle charts, raw geometric cancellation, disk-base absorption, and a simultaneous primitive-annulus rule. The latter uses a spare unattached base boundary to supply dual disks. The incoming hierarchy report proves a more general relative-band theorem for nonsaturated attachments and a one-two-handle localization for each saturated base. We use these results as starting points, give the needed geometry explicitly, and identify our additions.

Source checkpoint	Existing result	Role in this report
Maintained core cancellation	Saturated positive pivots; complete square row-leaf test; rectangular capped planner	Replace repeated rectangular scans and strengthen the completeness analysis
Maintained primitive annuli	A spare base boundary gives a simultaneous geometric primitive family	Preserve this case and admit certain fully attached bases

Source checkpoint	Existing result	Role in this report
Incoming hierarchy kernels	Relative one-handles for nonsaturated bases; one residual two-handle for each saturated base	Compute the residual old incidences and eliminate all added even columns from search
Incoming disconnected tile kernels	Exact finite kernels for repeated disconnected cubic tiles	Complementary terminal-pattern work; not rederived
Incoming moment-coded returns	Two-moment recovery of ordered boundary first returns	Complementary interface optimization; not integrated here
Incoming sparse cut grammars	Composition of bounded-cut graph responses through connected grammars	Complementary hierarchy-state work; not rederived

The current min-fill rectangular planner deserves a correction, rather than a false coverage claim. Its documentation calls it incomplete. Our theorem shows that its particular zero-fill priority is already complete for full raw row cancellation. The new algorithm therefore gives a simpler proof, smaller search work and faster replay; it does not create previously missing successes in that old raw model. New geometric coverage comes from the saturated-bundle extension.

1.3 The quasi-polynomial target and current primary sources

The 109-page 2021 talk linked by the repository states a running time $n^{O(\log n)}$[4]. The shorter Oxford handout displays $2^{O((\log n)^3)}$[5]. Both are quasi-polynomial, but they are distinct bounds. We retain the stronger $n^{O(\log n)}$ expression as the user's target and do not equate it with the handout's exponent.

Lackenby's July 2026 preprint[6] presents a complete hierarchy algorithm and efficient certificates, while reserving speed-up for later work. Its Section 9 explicitly does not provide the precision needed for a running-time estimate. Proposition 9.1 bounds iterations in terms of hierarchy length L and pattern-complexity bound g by $L(g+1)^L$. Theorems 14.1 and 14.4 concern efficient certification and compressed cut structure. These results give important mathematical infrastructure, but they do not turn an arbitrary local optimization into the announced global time bound. Section 8 records the remaining obligations and the exact place of our operator.

1.4 Prior art and the intended novelty claim

Unit-column peeling and stopping sets have substantial prior art. Richardson–Urbanke[7, Appendix C] define stopping sets on constraint rows, prove union closure, and identify the unique maximal stopping set as the residual of degree-one-column elimination. Di et al.[8, Definition 1.1 and Lemma 1.1] use the corresponding criterion for erasure decoding, including a multigraph formulation. Uniquely restricted matchings provide a related matrix language[9]. We do not claim to invent these constructions.

The mathematical additions developed here are the explicit persistence identity under (1), its complete geometric raw-cancellation consequence, the source-incidence formula for saturated residual curves, and the resulting parity exclusion of unexpanded band handles. The code contribution is a native producer and separate replay path for these results. The asserted novelty is relative to

the inspected project and the primary sources just identified. No claim of exhaustive bibliographic priority is needed for the proofs or implementation.

1.5 Notation and conventions

All matrices have entries in $\mathbb{N} = \{0, 1, 2, \ldots\}$. Counts are unsigned geometric counts, not exponent sums. A row denotes an attaching curve or a residual saturated-bundle relation, and a column denotes a one-handle cocore. We allow zero rows, empty matrices and unused columns. A full cancellation removes all rows; it need not remove all columns.

For a source triangulation, T denotes its tetrahedron count, P its number of active interface ports, and B the aggregate bit-length parameter of its binary coordinates and topology counters. For a finite sparse matrix, m, n, I denote its row count, column count and number of nonzero entries. Distinguishing I from the sum of numerical entries is essential: neither large multiplicities nor large base genus are expanded.

2 Stopping sets survive positive raw cancellation

Definition 2.1 (Positive raw pivot)**.** Let $N = (n_{ih})$ be a nonnegative integral matrix. A positive raw pivot at (r, g) is allowed when $n_{rg} = 1$. It deletes row r and column g and replaces the other entries by (1). A full raw cancellation is a finite sequence of such pivots that removes every row.

The adjective positive refers to the plus sign in the update. In particular, this operation is not Gaussian elimination over a field. It records a geometric disk-slide model before any subsequent reduction of intersections.

Definition 2.2 (Stopping set)**.** A nonempty set R of rows of N is a stopping set if, for each column h,

$$S_h(R) := \sum_{i \in R} n_{ih} \in \{0, 2, 3, \ldots\}. \tag{2}$$

Thus no column has exactly one crossing into R. The empty set is admitted when discussing unions, but it is never an obstruction certificate.

A coefficient two in a single row counts as two crossings. Replacing the matrix by its Boolean support would lose precisely this distinction.

Theorem 2.3 (Persistence under every positive pivot)**.** *Suppose R is a stopping set of N and a positive raw pivot is performed at (r, g). If $r \notin R$, then R is a stopping set of the new matrix. If $r \in R$, then $R \setminus \{r\}$ is nonempty and is a stopping set of the new matrix. Consequently a matrix with a nonempty stopping set admits no full raw cancellation.*

Proof. First let $r \notin R$. For every surviving column h,

$$S'_h(R) = S_h(R) + S_g(R) n_{rh}.$$

Each summand is either zero or at least two, because $S_h(R)$ and $S_g(R)$ have that property. Their sum therefore cannot equal one.

Now let $r \in R$. Since $n_{rg} = 1$, the stopping condition at g forces $S_g(R) \geq 2$. Put

$$k = \sum_{i \in R \setminus \{r\}} n_{ig} = S_g(R) - 1 \geq 1.$$

In particular $R \setminus \{r\}$ is nonempty. For any surviving h,

$$\begin{aligned} S'_h(R \setminus \{r\}) &= S_h(R) - n_{rh} + k n_{rh} \\ &= S_h(R) + (k-1) n_{rh}. \end{aligned} \tag{3}$$

If $S_h(R) = 0$, nonnegativity implies $n_{rh} = 0$ and the expression is zero. If $S_h(R) \geq 2$, it remains at least two. This proves persistence in both cases. Inductively, a nonempty stopping set survives every prefix of a positive pivot sequence, so such a sequence cannot remove every row. □

The short identity (3) is useful because it rules out *all* positive pivot orders directly. It does not presume a matching, a chosen survivor set or a deterministic producer.

Definition 2.4 (Column-leaf deletion)**.** A column g is a unit leaf in the current matrix if $\sum_i n_{ig} = 1$. There is then a unique incident row r, its entry is one, and its positive pivot has no fill: all surviving n_{ig} vanish. A column-leaf deletion removes this row and column.

Theorem 2.5 (Complete rectangular raw cancellation)**.** *For every finite nonnegative integral matrix N, the following are equivalent:*

(i) N admits a full positive raw cancellation;

(ii) N has no nonempty stopping set;

(iii) every maximal sequence of unit column-leaf deletions removes all rows;

(iv) some sequence of unit column-leaf deletions removes all rows.

In particular, deciding full raw cancellation requires no branching over pivot orders or over the columns that will survive.

Proof. Theorem 2.3 gives (i)⇒(ii). Run any maximal column-leaf sequence. If rows remain, call their set R. Every undeleted column has total zero or at least two on R, because the sequence is maximal. Each deleted column had only its designated row live when it was removed; hence its total on the final R is zero. No fill has ever occurred. Therefore R is a stopping set of the original matrix. This proves (ii)⇒(iii). Implication (iii)⇒(iv) is immediate, and every leaf deletion is a valid raw pivot, giving (iv)⇒(i). □

Corollary 2.6 (A canonical obstruction)**.** *There is a unique largest stopping set, with the empty set allowed. Every maximal column-leaf sequence leaves exactly this row set, denoted* $\mathrm{Stop}(N)$*. Thus the residual row set is independent of tie choices, although pivot columns and successful schedules need not be unique.*

Proof. The union of stopping sets is a stopping set: if a column meets the union, it meets one constituent set and already has total at least two there. Because there are finitely many rows, their union gives the largest set. No column leaf can delete a row in an extant stopping set, since that set would have column sum one. Hence every original stopping set lies in the final residual. The proof of Theorem 2.5 shows that the residual is itself a stopping set. The two containments prove the claim. □

This canonical peeling statement is the classical stopping-set phenomenon identified in the prior-art discussion. The additional persistence theorem explains why the same obstruction remains valid when the allowed operations are enlarged from leaf deletions to all positive raw pivots.

2.1 Exact saturation and sparse representation

Lemma 2.7 (Saturation at two is exact for the model)**.** *The map* $\mathrm{cap}_2(a) = \min(2, a)$ *preserves addition and multiplication when the target operations are capped at two. It preserves the singleton predicate and the stopping-set predicate. It therefore preserves the existence of a full positive raw cancellation.*

Proof. For $a, b \in \mathbb{N}$, both $\min(2, a + b) = \min(2, \min(2, a) + \min(2, b))$ and $\min(2, ab) = \min(2, \min(2, a) \min(2, b))$ follow by splitting into whether the result is zero, one or at least two. The same identities commute with (1). A sum is one exactly when one summand is one and all the others are zero, which is also unchanged by capping. □

Actual crossing multiplicities may consequently be arbitrarily large in binary without changing the finite peeling state. Values zero, one and at least two are all that the decision needs. This statement does not authorize replacing a geometric count by a signed exponent sum.

2.2 A matching interpretation, without a matching algorithm

Corollary 2.8 (Unit uniquely restricted row matching)**.** *A full raw cancellation exists if and only if there is a set of* m *columns whose* $m \times m$ *submatrix has permanent one. Equivalently, the weighted bipartite incidence multigraph has a row-saturating matching whose induced square subgraph has a unique perfect matching, all of whose chosen edge multiplicities are one.*

Proof. Suppose peeling removes all rows. Select the m pivot columns and order both rows and those columns by deletion time. A pivot column has its diagonal entry one and no nonzero in any later row. Thus the selected square matrix is triangular with unit diagonal. Its only nonzero permanent term is the diagonal term.

Conversely, a nonnegative integral square matrix of permanent one has a unique perfect matching with unit chosen entries. Orient an off-matching incidence from its row's matched pair to the incident column's matched pair. A directed cycle would give an alternating cycle and a second perfect matching. The directed graph is therefore acyclic and has a source. Its matched column has no incidence from other rows. Removing that pair and repeating gives column-leaf deletion of the selected minor. These are also column leaves in the full matrix, because the selected minor includes every row. Other columns do not affect a leaf's column sum. □

There is no need to compute a permanent or to enumerate the selected columns. The direct obstruction theorem supplies the complete algorithm. The distinction between *saturating all rows* and maximizing the size of a partially constrained matching should be retained when comparing this corollary with matching literature.

2.3 Boundary cases and limits of the conclusion

Example 2.9 (A zero row)**.** A single all-zero row is a stopping set. A raw singleton cancellation cannot remove its two-handle, even if that handle has some other geometric disposal, such as a later two/three-handle cancellation. Three-handles are outside the present model.

Example 2.10 (Peeling is not maximum partial cancellation)**.** For $N = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, every column has total two and no column leaf exists. A positive pivot can nevertheless remove one row and leave the matrix (2). Thus $|\mathrm{Stop}(N)|$ is not the minimum number of rows attainable by an arbitrary partial raw sequence. The theorem decides complete cancellation and gives a canonical source obstruction; it does not optimize partial progress.

Example 2.11 (A unimodular raw obstruction)**.** The matrix

$$U = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$$

has determinant one, but its entire row set is a stopping set. Every legal first raw pivot leaves the single entry three. Signed linear algebra therefore cannot replace the unsigned criterion. The group presentation $\langle x, y \mid xy, xy^2 \rangle$ is trivial, since the two equations give $y = 1$ and then $x = 1$. Its raw occurrence matrix is U.

There is also a geometric ball realization. Begin with the standard genus-two handlebody and its two cancelling two-handles, attached along the two primitive generator curves. Slide the first attaching curve over the second to obtain xy, then slide the second over the first to obtain the cyclic word xy^2, choosing the bands with no additional cocore crossings. These are ordinary geometric two-handle slides, so the resulting manifold is still a ball. The indicated raw incidence is U. This supplies a concrete reason that a raw obstruction must not be reported as a nonball or knottedness certificate.

3 Sparse discovery, independent replay, and the old planner

3.1 An immutable-source algorithm

Peeling needs no evolving matrix. For every source column h, maintain the number d_h of incident live rows and the identity of its sole owner when $d_h = 1$. An incident row with entry at least two does not make a unit leaf. When row r is removed, visit its source nonzero columns once, decrease their incident-row counts, and enqueue newly exposed unit leaves.

Listing 1: Complete raw cancellation from source incidence.

```
build row-to-column incidence lists
for each column h:
    degree[h] = number of nonzero incident rows
    owner[h] = XOR of those row labels
enqueue h if degree[h] == 1 and N[owner[h], h] == 1

while the queue is nonempty:
    pop h
    if degree[h] != 1: continue
    r = owner[h]
    record the cancellation (r,h)
    mark r deleted
    for each nonzero source column c of row r:
        degree[c] -= 1
        owner[c] ^= r
        if degree[c] == 1 and N[owner[c],c] == 1:
            enqueue c

return the schedule and the remaining row set
```

XOR is only an efficient owner-recovery implementation. Dense row labels make it a word operation; an incident-list representation gives the same mathematical algorithm. An all-zero row is retained. A column that is initially absent is irrelevant. The queue may contain stale entries, which are ignored after their degrees cease to be one.

Theorem 3.1 (Sparse structural bound)**.** *After indexing the rows and columns, the complete raw-cancellation decision, a successful schedule or a canonical stopping set, and the associated sparse source data require $O(I + m + n)$ structural operations and $O(I + m + n)$ words of storage. A successful schedule has at most m pairs. If labels use $O(\log(m + n + 2))$ bits and the binary input length is Λ, a deterministic dense-array implementation has bit cost*

$$O(\Lambda + (I + m + n)\log(m + n + 2))$$

under elementary word arithmetic. Sorting arbitrary external labels may add a logarithmic preprocessing factor.

Proof. Each nonzero is read once during construction and visited at most once when its row is deleted. A column's live degree decreases monotonically, so it becomes degree one at most once. It is consequently enqueued at most once, including initial leaves. Queue processing is linear in the number of appearing columns. All other work is proportional to the row and column arrays or the schedule. Entries are capped at two while reading; the later algorithm does not perform arithmetic on their original magnitudes. Counters and row labels require logarithmic space. These observations give both structural and bit bounds. □

The provided core producer follows this sparse queue design. The standalone saturated-bundle producer calls the same public sparse planner. Python dictionaries are implementation data structures, not a proof of adversarial constant-time hashing. The deterministic bound above describes the indexed algorithm; the empirical measurements concern the supplied Python implementation.

3.2 Positive certificates without producer replay

Let a proposed schedule be $((r_0, g_0), \ldots, (r_{s-1}, g_{s-1}))$. Define $\tau(r_i) = i$ and $\tau(r) = +\infty$ for every unlisted row. No row or pivot column may repeat.

Theorem 3.2 (Timestamp certificate)**.** *The schedule is a valid no-fill column-leaf prefix if and only if, for each i, the source entry $n_{r_i g_i}$ equals one and every different row r with $n_{r g_i} > 0$ satisfies $\tau(r) < i$. It is a complete certificate if, in addition, every source row is listed. These conditions can be checked by one scan of the source incidence and the schedule.*

Proof. At step i, the live source rows are exactly those with deletion time at least i. The stated inequality therefore says that column g_i has only row r_i left and that its entry is one. Previous column-leaf pivots have created no fill, so the source incidence on live rows is the actual incidence. Induction proves validity. Conversely, each genuine leaf step forces those source incidences to belong to earlier deleted rows. The extra coverage condition is exactly complete cancellation. □

The independent checker does not call the producer, maintain its queue, or carry out (1). It builds row timestamps and checks source inequalities. The certificate occupies $O(m \log(m + n + 2))$ bits, apart from externally supplied source identifiers. The same checker can verify a proper prefix by retaining $+\infty$ on unlisted rows.

3.3 Obstruction certificates

An obstruction certificate supplies a nonempty, duplicate-free list R of source row labels. The checker accumulates $\min(2, \sum_{r \in R} n_{rh})$ for each appearing column and rejects if any result is one. Theorem 2.3 proves that an accepted certificate rules out all full positive raw schedules for those

input counts. The checker need not trust the producer's claim that R is canonical; any nonempty stopping set suffices.

This distinction is useful for independent verification. Canonical discovery is a deterministic algorithmic convenience, whereas the obstruction theorem is a short local certificate. Relabelling the source changes labels, not the existence of the certified stopping set. Changing the disk system can change the matrix and may remove its obstruction.

3.4 Why the retained capped planner was already complete

The source implementation's legacy capped planner scores a candidate $n_{rg} = 1$ by

$$\left(\sum_h n_{rh} - 1\right)\left(\sum_i n_{ig} - 1\right)$$

using capped entries. A zero score means that the donor is a unit row leaf or that the pivot column is a unit column leaf. In either case no surviving entry receives fill. The planner chooses a minimum score.

Proposition 3.3 (Completeness of zero-fill priority)**.** *A producer that always chooses a zero-fill raw pivot when one exists is complete for full positive row cancellation. In particular, the retained min-fill capped planner already had this completeness property, despite its earlier documentation.*

Proof. If the current matrix is fully cancellable, Theorem 2.5 supplies a unit column leaf, so a zero-fill pivot exists. A column-leaf pivot preserves full cancellability: any stopping set of its reduced matrix would already be a stopping set of the original matrix, since the removed pivot column is zero on the remaining rows.

Consider instead a unit row leaf r at column g. If the reduced matrix had a stopping set R, examine its source sum at g. If that sum is zero or at least two, R itself obstructs the original matrix. If it is one, then $R \cup \{r\}$ obstructs it: the sum at g becomes two and the row leaf contributes nothing elsewhere. Either case contradicts full cancellability. Hence this zero-fill pivot is safe as well.

Induction shows that zero-fill priority succeeds whenever any positive schedule succeeds. Its successful outputs are always valid schedules, so the converse implication requires no further argument. □

This proof does not assert that *every* arbitrary singleton choice preserves full cancellability. It establishes precisely the property used by the retained producer. The native change removes repeated candidate scans and unneeded positive updates, while preserving its complete-success set. Legacy certificates remain independently replayable.

3.5 A family exposing the avoidable scan cost

For $m \geq 2$, consider the $m \times (m+1)$ bidiagonal chain with $n_{r,r} = n_{r,r+1} = 1$ and all other entries zero:

$$N_m = \begin{pmatrix} 1 & 1 & 0 & \cdots & 0 \\ 0 & 1 & 1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & 1 & 1 \end{pmatrix}.$$

It has $I = 2m$ source incidences, no initial unit row leaf, and exposed unit columns at the ends. Peeling successively exposes the next column. A planner that rescans every surviving incidence at

each step performs $\Theta(m^2)$ source visits. The queue algorithm performs $\Theta(m)$ visits. Both find a complete certificate; their distinction is avoidable repeated work.

This is a matrix family, not a benchmark of knot recognition. It has a geometric interpretation as a primitive attaching family on a handlebody with one remaining handle, but an arbitrary caller-provided copy of the matrix does not authenticate that interpretation. The genuine source-cut audit in Section 7 supplies the separate geometric validation for the native paths.

3.6 An exact boundary between useful and wasted raw search

After a verified stopping set is found, further enumeration of positive pivot orders cannot establish complete cancellation. This is a mathematical termination rule for the raw subproblem, not a heuristic timeout. A higher level search must change something outside this model: the geometric disk system, the forest, the attaching geometry, or the decomposition surface.

For this reason the residual row set is a useful search diagnostic. It names the curves whose joint incidence has no unique crossing. It does not measure all remaining topology, and its cardinality is not a global hierarchy potential. A strategy that treats the residual size as a guaranteed decreasing topological complexity would need a further theorem.

4 Exact transport of additional incidence counts

Complete no-fill schedules also simplify the transport of an additional nonnegative incidence row. This is an algebraic refinement with a clear use in later geometric work. It transports counts under the stated positive update; it does not recover cyclic order, strand endpoints or an embedding.

Let N now contain the *exact uncapped* source counts, and let $((r_0, g_0), \dots, (r_{m-1}, g_{m-1}))$ be a verified complete column-leaf schedule. Let $p = (p_h)$ be an additional nonnegative row, called a passenger. When the pivot (r_i, g_i) is applied to it, the formal positive transport rule is

$$p'_h = p_h + p_{g_i} n_{r_i h} \quad (h \neq g_i), \qquad p'_{g_i} \text{ is deleted.} \tag{4}$$

In a geometric application this formula must itself be justified by the chosen disk slides and the placement of the passenger curves. The theorem below is conditional on this exact update rule, not on abelianized words.

Lemma 4.1 (Immutable donors form a directed acyclic graph)**.** *For $j < i$, the source entry $n_{r_i g_j}$ is zero. A donor row therefore contains only its own pivot column, later pivot columns and survivor columns. Apart from deleting its pivot entry, its exact source coefficients can be used without any row update throughout the whole schedule.*

Proof. When column g_j was used, row r_i was still live. The column-leaf condition forces its entry in that row to be zero. Every preceding pivot has no fill in any surviving donor, so the surviving row entries have always agreed with the source. This proves both statements. □

Form a directed weighted graph on the matrix columns. For every nonpivot entry of donor row r_i, put an edge $g_i \to h$ of weight $n_{r_i h}$. Order the pivot columns by schedule time and put survivors afterwards. Lemma 4.1 makes this a directed acyclic graph, with survivor columns as sinks. The pivot entries themselves do not give edges.

Theorem 4.2 (Sparse positive transport and a bit bound)**.** *One passenger row can be transported exactly by reading every nonpivot donor incidence once, using $O(I + m + |\operatorname{supp} p|)$ big-integer operations. For a survivor h, the final entry is*

$$p_h^{\mathrm{out}} = \sum_v p_v \sum_{\gamma: v \rightsquigarrow h} \prod_{e \in \gamma} w_e, \tag{5}$$

with the length-zero path included. Let E be the number of donor graph edges, let

$$L_N = \sum_e \lceil \log_2(1 + w_e) \rceil, \qquad B_p = \max\left(\{0\} \cup \{\lceil \log_2(1 + p_v) \rceil : v \in \operatorname{supp} p\}\right),$$

and let q be the number of appearing source or passenger columns. Every intermediate and final nonzero passenger entry has bit length at most

$$B_p + L_N + E + \lceil \log_2(q + 1) \rceil + 1. \tag{6}$$

Hence transport is polynomial in the encoded exact source data, even when the numerical coefficients grow exponentially in the number of rows.

Proof. At pivot g_i, distribute its current passenger coefficient along its outgoing donor edges, multiplying by their weights, and then delete it. The immutable-donor lemma shows that these are exactly the updates (4). Every graph edge is processed once, giving the structural operation count.

Induction in topological order identifies the value reaching a vertex with the sum over all directed paths reaching that vertex. This proves (5); columns occurring only in the passenger have only their length-zero contribution.

A directed path uses every edge at most once. Its edge set determines the path, so there are at most 2^E paths from any fixed starting vertex to a specified vertex. Each path product is at most 2^{L_N}, and each initial passenger value is less than 2^{B_p}. Summing over at most q starting vertices gives the stated conservative bit bound. An intermediate entry is a sub-sum of the analogous nonnegative path expression ending at that vertex, so it satisfies the bound as well. Polynomial-time integer multiplication then gives polynomial bit complexity. □

Example 4.3 (Two branching paths)**.** Use columns (g_0, g_1, g_2, z) and the source rows

$$N = \begin{pmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 7 \end{pmatrix}, \qquad p = (11, 0, 0, 13).$$

The complete column schedule uses g_0, g_1, g_2. The final survivor entry is

$$p_z^{\mathrm{out}} = 13 + 11(2 \cdot 5 + 3 \cdot 7) = 354.$$

Using capped donor counts would instead give 101. Capping is exact for the cancellation *decision*, but not for the transported values.

The delivered `raw_passenger_transport` module first checks the full schedule using the independent source timestamp verifier, then applies the uncapped donor updates. Its tests compare with an independent explicit positive-pivot oracle. The API includes columns that appear only in the passenger and retains them as survivors.

This gives a concrete polynomial-bit algebraic step toward tracking additional curves through a successful cancellation. It is intentionally short of geometric boundary-pattern transport. Two patterns with identical unsigned incidence can have different cyclic orders and topology. Moreover, the virtual band columns of Section 5 were omitted for a decision proof; this theorem does not reconstruct per-band passenger data from a genus counter. A compressed geometric representation of those additional markings remains a research problem.

5 Saturated interval bundles and joint geometric cancellation

The existing primitive-annulus construction[1] applies when a bordered base surface has a boundary circle that is not fully attached to the core. We now remove that restriction under a different, explicitly checkable condition. An entire saturated bundle contributes one additional two-handle relation. We cancel these relations jointly with the original core two-handles, using only the original core cocores. The genera of the bases can remain in binary throughout.

Here a *saturated* bundle is an orientable total interval bundle $P_i \to F_i$ over a connected compact surface with nonempty boundary, all of whose vertical boundary annuli are attached to the core. The base F_i need not be orientable. We work with actual embedded annuli with disjoint collars; an abstract cyclic word is insufficient to establish this geometric hypothesis. In the normal-cut application the source parallelity construction and its ordered boundary observer supply these annuli and collars.

The incoming hierarchy report[3] already proves the one-two-handle count used below. The new content of the relative lemma is the exact old-cocore incidence formula and its implications for excluding all unexpanded band columns from the raw search.

5.1 One relative relation, with exact source incidences

Let U be a disjoint union of handlebodies given by zero- and one-handles. Let $\alpha_1, \dots, \alpha_f$ be the pairwise disjoint attaching curves of the original core two-handles. The core has no three-handles. Fix a spanning forest in the original zero/one-handle graph, and let $\{D_h : h \in J\}$ be the cocore disk system corresponding to the remaining one-handles. Each saturated bundle is attached along annuli disjoint from the α_r. Write β_{ij} for an actual embedded core curve of its jth attaching annulus, viewed on the U side.

All curves are transverse to the cocore boundaries. Define

$$a_{rh} = |\alpha_r \cap \partial D_h|, \qquad a_{ih}^{\mathrm{sat}} = \sum_{j=1}^{b_i} |\beta_{ij} \cap \partial D_h|, \tag{7}$$

where $b_i = |\pi_0(\partial F_i)|$. These are unsigned geometric counts. Oppositely oriented passages both contribute, and no free reduction is performed.

Lemma 5.1 (Relative bundle presentation with exact incidences)**.** *The union of U with the saturated bundles has a relative handle decomposition with, for each P_i, precisely*

$$h_i = 1 - \chi(F_i)$$

new one-handles and one new two-handle, and no new zero- or three-handles. The new two-handle for P_i has incidence a_{ih}^{sat} with every old cocore D_h. It traverses each of its own new one-handles exactly twice and every other bundle's new one-handles zero times. Original core two-handles traverse all the new one-handles zero times.

Proof. Choose pairwise disjoint properly embedded arcs in F_i cutting it into one disk. Their number is $1 - \chi(F_i)$: each cut increases Euler characteristic by one. Equivalently, F_i is obtained from collars of all its boundary circles by attaching h_i bands and then one final disk. This relative surface handle decomposition exists for orientable and nonorientable bordered surfaces alike.

Thicken this construction using the given interval bundle. A boundary collar can be absorbed into a collar of its annular attachment site in U. Each band thickens to a three-dimensional one-handle,

and the final disk, whose interval bundle is trivial, thickens to a two-handle. This proves the asserted handle counts.

The boundary of the final disk follows each long side of each band once. Its attaching curve therefore crosses every new band cocore exactly twice. A twisted band can change the signs of those crossings, but cannot change this unsigned count. The same boundary follows each original boundary segment of F_i once. Those segments partition the original boundary circles, so every original boundary circle is followed once in total. A whole boundary circle need not occur consecutively in the attaching word. Counting its segments avoids choices of basepoints or conjugating paths.

For the exact counts on U, first straighten the annular attaching collars about their chosen core curves. A relative Dehn twist of a vertical annulus extends across a collar in the interval bundle: in collar coordinates it has the form $(\theta, r, t) \mapsto (\theta + 2\pi k t \rho(r), r, t)$, where ρ is one at the vertical boundary and zero at the inner edge. Its effect on the horizontal boundary is allowed because the conclusion here is unmarked. Thus annular gluing parameters introduce no additional restriction on the positions of the band feet.

Now narrow each annular attaching collar about its chosen core curve β_{ij}, fixing that core curve. In transverse position, the intersections with the cocore boundaries then consist of short spanning arcs, one for each counted intersection with β_{ij}. The finite band feet can be chosen in strips avoiding all these arcs. Consequently the old-boundary portions of the final attaching curve contain every counted crossing exactly once, and the new bands and their feet introduce no further crossing of an old cocore. This proves (7). The collar changes are supported away from the original two-handle attaching curves and the other marked interfaces.

Finally, every added band belongs to its own bundle. No original two-handle or other bundle's final disk uses it, giving the remaining zero incidences. □

For a disk base, $h_i = 0$, and the lemma is exactly the familiar absorption of a fully attached disk bundle as a two-handle. An annulus and a Möbius band both have $h_i = 1$. Their additional one-handle occurs twice in the residual relation; its orientation data can differ, but the incidence argument is the same.

5.2 Why the additional columns can be omitted completely

Let A be the matrix obtained by stacking the f original core rows and one row a_i^{sat} per saturated bundle, keeping only the old nonforest columns J. In particular, a bundle contributes one row even when h_i is very large.

Extend the fixed original forest to a forest for the enlarged zero/one-handle graph using only added band edges. This is possible because the original forest already spans each original core component. Contraction deletes the selected added columns and leaves all old nonforest incidences unchanged. Let N denote the full remaining matrix, including the added band columns that were not contracted.

Theorem 5.2 (Exact reduction to the aggregate source matrix)**.** *The full relative matrix N admits a complete sequence of positive geometric singleton pivots if and only if A does. Thus the source column-leaf algorithm decides complete cancellation in this selected relative handle decomposition without constructing any added band column.*

Proof. Every entry in an added band column is initially even, by Lemma 5.1. Recall that a positive pivot (r, g), with $n_{rg} = 1$, updates surviving entries by

$$n'_{ih} = n_{ih} + n_{ig} n_{rh} \qquad (i \neq r,\ h \neq g).$$

If column h has only even entries, its updated entries are even as well. Inductively, no added band column can ever contain a singleton. Every possible pivot must therefore use an original core column.

Restricted to the original core columns, each update is precisely the same update applied to A. A complete sequence for N consequently gives one for A. Conversely, a complete sequence for A can be applied to N with the same pivot entries, deleting all its two-handle rows. Theorem 2.5, which equates positive singleton cancellation with source column-leaf deletion, completes the argument. □

In fact, evenness is stronger than the algebra needs. A column whose entries are all zero or at least two can never acquire an entry one under a positive update: both summands are zero or at least two. Thus any such column can be omitted when deciding full raw cancellation. The geometric band construction gives the particularly transparent values zero and two.

The qualifier "selected relative handle decomposition" matters. The theorem permits every positive pivot in that decomposition, not arbitrary changes of disk system or arbitrary handle moves. An uncancelled stopping set is an obstruction to this rule, not an obstruction to the underlying manifold being a handlebody.

Corollary 5.3 (Joint core and saturated-bundle certificate)**.** *If source column-leaf deletion removes all rows of A, then the core with all saturated bundles attached is a disjoint union of handlebodies. The cancellation order may interleave original core two-handles and saturated-bundle relations.*

Proof. Every pivot has the geometric realization of a one/two-handle cancellation. For a source column leaf, no surviving attaching curve meets the pivot cocore, so the positive update has no fill in any surviving row. Lemma 5.1 authenticates the geometric rows, and Theorem 5.2 shows that the omitted columns hide no extra two-handle relation. After all rows have been removed, only zero- and one-handles remain. Each connected component is therefore a handlebody. □

5.3 A primitive anchor explains the geometry

The following special case makes the saturated-bundle step intuitive without introducing its full relative matrix. In this proposition, the remaining marked interfaces mean the other attachment sites on H and the displayed vertical annuli. No arbitrary graph on the horizontal boundary of P is included in that phrase.

Proposition 5.4 (One primitive anchor releases a saturated bundle)**.** *Let H be a handlebody and let $P \to F$ be an orientable interval bundle over a connected bordered surface. Choose one vertical annulus A_0. Suppose its attachment annulus on H has a properly embedded dual disk E meeting that annulus in one spanning arc and avoiding all other marked interfaces. After attaching P to H only along A_0, the result is a handlebody of genus*

$$g(H) + 1 - \chi(F) - 1 = g(H) - \chi(F).$$

All remaining vertical annuli of P form a simultaneous primitive family in this handlebody, relative to the remaining marked interfaces.

Proof. For the first annular gluing, cut the annulus along a spanning arc to obtain a disk interface. Restoring the annulus supplies a connecting one-handle and a two-handle. The latter crosses the handle dual to E exactly once, so that pair cancels geometrically. The pieces are then joined by the connecting handle, giving the claimed genus. This is the usual primitive-annulus argument[10]; the explicit dual-disk construction is the hypothesis used here.

For every other boundary circle C_j of F, choose a properly embedded arc γ_j joining it to C_0, with the arcs pairwise disjoint. Such arcs can be chosen successively by joining distinct boundary components; cutting along one leaves the base connected and merges those two boundary components. This construction does not require orientability of F.

Each $p^{-1}(\gamma_j)$ is a rectangle with one edge on A_0 and one edge on the annulus over C_j. Choose pairwise disjoint parallel copies E_j of E. A collar isotopy matches their spanning arcs in A_0 with the corresponding rectangle edges: any finite disjoint family of spanning arcs of an annulus is parallel. Gluing each rectangle to its E_j along that edge gives an embedded proper disk in $H \cup_{A_0} P$. These disks are pairwise disjoint. Each meets its designated remaining annulus in one spanning arc and avoids all the others, proving simultaneous primitivity. □

A singleton in an aggregate saturated row supplies exactly such an anchor after the preceding source cancellations. The relative-handle proof is more general: anchors need not exist simultaneously in the original core and may become available only after other rows disappear.

5.4 Completing the census and its complexity

After the joint cancellations, attach every unsaturated bundle using the established spare-boundary primitive-annulus construction, and attach the remaining proper disk interfaces. These operations yield handlebodies. They can be delayed until after the joint cancellation because their disjoint marked sites persist under the resulting homeomorphisms. No explicit coordinates for their transported images are required for this unmarked topological conclusion.

Let C be a connected component of the full attachment graph. Its Euler characteristic is

$$\chi_C = \sum_{K_v \subset C} \chi(K_v) + \sum_{P_i \subset C} \chi(F_i) - d_C, \qquad g_C = 1 - \chi_C, \tag{8}$$

where d_C is the number of proper disk interfaces in C; annuli have Euler characteristic zero. Equation (8) computes the genus only after the cancellation and gluing proof establishes that C is a handlebody. It cannot be used as a stand-alone ball test. An unported bordered bundle is an isolated handlebody of genus $1 - \chi(F_i)$ and retains its full binary multiplicity.

For the saturated subassembly alone, suppose a connected component contains k original zero/one-handlebody components, of genera g_v, and f_C original core two-handles. The same count becomes

$$g_C = \sum_v g_v - f_C - k + 1 - \sum_i \chi(F_i).$$

It is independent of how many added band edges were needed to join the original components.

Let r be the number of aggregate and original rows, c the number of old nonforest columns, and I the number of nonzero source incidences. After source reconstruction, the queue-based column-leaf algorithm uses $O(I + r + c)$ structural operations; a heap for a deterministic least-label order uses $O(I + (r + c)\log(r + c + 1))$. Counts can be saturated at two, because only zero, one, and at least two are relevant. The certificate lists at most r source row/column pairs and can be checked by one incidence scan together with deletion times. Base Euler characteristics, genera, and isolated multiplicities are manipulated in binary. No operation is proportional to $\sum_i (1 - \chi(F_i))$.

The complete normal-cut wrapper must additionally authenticate the normal vector, compressed parallelity profiles, original handle charts, and cyclic annular interface words. Its total bound includes those existing source routines and their binary arithmetic. Closed base surfaces and core three-handles remain outside this certificate. Failure of the source cancellation test returns an inconclusive result.

This extension proves the unmarked homeomorphism type of more supplied cuts. It supplies neither an inherited boundary-pattern encoding for the next hierarchy stage nor a bound on the number of candidate surfaces or hierarchy resets. Those are the remaining global issues that a quasi-polynomial unknot-recognition argument must address.

6 Native implementation and certificate boundaries

6.1 Core cancellation integration

The implementation lives in `Topology/UnknotRecognition/fast`. The successful row-leaf fast path is retained. For a rectangular core, `core_handlebody_certificate` now calls `column_cancellation_plan`, which scans immutable source incidence and returns a schedule together with its residual rows and structural counters. The old `capped_cancellation_plan` remains available for compatibility and paired comparisons; its completeness documentation is corrected as in Proposition 3.3.

The new module `column_cancellation_verify` supplies two checks:

- `verify_column_cancellations` checks deletion timestamps.
- `verify_raw_cancellation_obstruction` checks the stopping condition directly.

Neither imports a producer. The core verifier supports a nested `normal-core-handlebody-v2` certificate with method `column_leaf`. Existing `normal-core-handlebody-v1` leaf and capped certificates remain accepted. The surrounding whole-cut certificate schemas need not change simply because their nested core proof is faster.

The old square row-leaf obstruction path remains a specialized early exit. The canonical stopping-set theorem applies directly to the public column planner's returned residual; a legacy row-leaf residual is not silently relabelled as that canonical source set. Callers requiring a new independent stopping certificate should use the column planner and the stopping checker.

Linear cancellation complexity is a statement about the kernel after source incidence has been built. Existing forest validation, source geometry reconstruction and component assembly keep their own costs. We do not claim that modifying the planner makes those unrelated operations linear.

6.2 Source compilation for saturated bases

The new `normal_cut_saturated` module adds a separate whole-cut query. Its low-level finite operation is `saturated_source_census`; the public wrapper is `normal_cut_saturated`. These are opt-in APIs. They do not dispatch a new verdict in the main knot recognizer.

The finite source compiler performs the following operations.

1. Choose a forest in the actual core zero/one-handle graph and recover the core components. A core three-handle triggers an inconclusive result.
2. Read the authenticated base profiles and ordered attachment components. Reject closed bases. Each ported base must have multiplicity one, and each maximal attachment must have a unique base owner and a unique core owner.
3. Count complete attaching circles for each base. A base is saturated exactly when this count equals its total number of boundary components. It then has no proper interval attachments. Unported boundary circles are included in the profile count, so they cannot be accidentally ignored.

4. Extract the old two-handle rows from the actual cyclic source flags. For each saturated base, traverse all of its full annular words and sum their old nonforest one-handle crossings into one row.
5. Apply complete column cancellation to the combined source matrix. If it succeeds, compute connected attachment components and their Euler characteristics, then use the proved handlebody class to obtain genera.

The fourth step needs more than a word of core vertex labels. An individual core one-handle may be a loop, so its two ends can belong to the same core zero-handle. The compiler checks the tetrahedron/face flags at both ends of every face-sector port, retains the direction of the corresponding midpoint path, and checks that consecutive path ends join to a closed curve. Every crossing is counted in absolute value, including repeated passages and passages with opposite orientations. The independent verifier reconstructs the same geometry with a separate path routine.

Proposition 6.1 (Size of the compiled cancellation kernel)**.** *Let the old core have f two-handles and c nonforest one-handles, and let s bases be saturated. The compiled matrix has $f + s$ rows and c possible columns. If I_0 is the number of nonzero old core incidences and P is the total number of active ports, it has at most $I_0 + P$ nonzeros. In particular, its size is independent of the numerical sum $\sum_i(1 - \chi(F_i))$ of added band handles.*

Proof. There is one row per old two-handle and exactly one per saturated base. Only old nonforest columns are stored. Each full annular boundary word uses source ports; the attachment skeleton partitions the ports among maximal attachments. A core one-handle crossing uses such a port, and summing crossings into rows can merge nonzeros but cannot create more than the number of contributing ports. Theorem 5.2 shows that no omitted band column is needed for the cancellation decision. □

In the repository's source representation the finite core and active ports are bounded in terms of T[1]. Even using the coarse $I_0 \leq fc$, this is a polynomial-size kernel. With explicit source incidence lists, construction is charged to their actual total length. Binary Euler characteristics and multiplicities are retained as integers and use the established hexadecimal codec for transport.

6.3 The whole-cut proof and its independent replay

A successful wrapper returns a `normal-cut-saturated-v1` certificate with three components: the parallelity source proof, the ordered-boundary source proof, and a `saturated-source-census-v1` finite proof. The finite proof names the forest, every saturated base and its relation label, the full cancellation schedule, and the resulting census.

The independent `normal_cut_saturated_verify` module first replays both existing source proofs. Only then does it inspect the source geometry, base profiles and boundary skeleton contained in them. It independently checks the forest, reconstructs saturated ownership and raw crossing counts, checks every pivot, and computes attachment components by graph traversal. The producer uses a different component-construction path. Both check the redundant source Euler identity

$$\chi(\text{cut}) = \chi(\text{core}) + \chi(\text{bundles}) - \chi(\text{interfaces}).$$

That identity is an arithmetic consistency check after the geometric proof; it is not the proof that a component is a handlebody.

The supplied finite saturated verifier deliberately retains a direct scan of surviving source rows at each pivot. Its matrix-check cost is polynomial, with a coarse $O(mI + m^2)$ bound, rather than

the linear timestamp bound already implemented in the separate core checker. This keeps the new geometric replay easy to inspect. Theorem 3.2 identifies a later compatible optimization if this finite replay becomes material. Whole-cut cost still includes the source orbit proofs, which dominate many of the measured small examples.

There are six source orbit traces across the parallelity and boundary queries. The producer's cycle allowance is shared across the two stages; the wrapper subtracts the consumed parallelity cycles before the boundary query. Replay counts all six supplied traces against its operation cap. Caller abort exceptions propagate. Certificate mutation checks reject invalid schema fields, repeated pivots, nonexistent or Boolean labels, changed relative-handle counts, altered source charts and changed genus claims. The existing exact integer codec handles large signed and hexadecimal quantities.

6.4 Examples the extension must distinguish

A saturated disk. Here $\chi(F) = 1$, so the relative bundle has no one-handles and one two-handle. Its row is the full annular attachment row. The new construction recovers the existing disk-base absorption case.

A saturated punctured surface with a core anchor. Let a once-bordered orientable genus-q surface be attached to a solid torus along a core annulus meeting its cocore once. The aggregate matrix is (1), for every q. The old core handle cancels the residual two-handle, leaving a genus-$2q$ handlebody. The matrix size is constant even if q has thousands of bits. A nonorientable base of crosscap number q gives the same calculation with a genus-q result.

Two individually primitive annuli on the same handle. Take an annulus base and attach both boundary annuli to longitudinal annuli of one solid torus. The aggregate row is (2), so the criterion refuses. Treating the two boundary circles as separate singleton rows would be erroneous. With product gluings, this construction is the product of a circle with a disk to which one orientable band has been attached; the result is $T^2 \times I$, not a handlebody. Its fundamental group is $\mathbb{Z}^2$. The aggregate row detects the missing relative primitivity.

A saturated Möbius base attached with multiplicity three. The vertical annulus of the orientable interval bundle over a Möbius band winds twice around its solid-torus core. Gluing it to a winding-three annulus in another solid torus gives the amalgam $\langle x, y \mid x^3 = y^2 \rangle$, the familiar torus-knot group used in the incoming counterexample[3]. The old aggregate row (3) is blocked. Its infinite cyclic abelianization does not justify a handlebody conclusion. With a winding-one attachment instead, the row is (1) and the core generator cancels.

Interleaved core and bundle relations. The combined matrix need not be solved in two phases. A saturated row can expose the next old core row, and an old core cancellation can expose a saturated row. The producer handles this mixed schedule directly. If the combined matrix fully cancels, its original-core row submatrix also fully cancels: any stopping set of that submatrix would obstruct the combined matrix. Thus joint cancellation cannot cure a fixed old-core raw obstruction.

The distinction concerns the choice and retention of a compatible schedule. For the algebraic matrix

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \end{pmatrix},$$

declare the first row to be an old core relation. The joint column schedule pivots first in its third column, then in the second row's second column. But the core-only matrix also admits a pivot in its first column. Committing to that choice transports the second row to $(3, 2)$, which is blocked. An arbitrary core-only success and its unmarked census therefore do not replace a schedule checked against all annular data. This example asserts an algebraic scheduling fact, not an authenticated source realization of that matrix.

6.5 Exactly what a successful result certifies

The output is a complete binary histogram of handlebody genera for a *supplied normal cut*. Isolated bundle components contribute their full multiplicities. The result distinguishes one ball from several balls and from a ball accompanied by positive-genus components.

A knot conclusion needs the separate facts that the input is the authenticated exterior of the queried diagram and that the cutting data supplies the appropriate disk or hierarchy argument. The wrapper itself proves neither of those facts. In particular, a raw stopping certificate proves failure of a restricted cancellation model, while a resource-limited or unsupported query returns `INCONCLUSIVE`. These are part of the mathematical input/output contract, not optional interpretations of the results.

7 Validation and measured performance

The evidence has three different roles. Exhaustive small matrices test the algebra against an independent search over all permitted pivots. Genuine normal cuts test source compilation and the geometric census. Paired timing experiments measure the cost of particular implementations. None of these finite experiments proves the general theorems; their proofs appear above. The exact inputs, raw samples, logs and source hashes are supplied with this article, as described in Appendix A.

7.1 Algebraic and certificate validation

The column-cancellation tests enumerate every legal saturated positive-pivot schedule on 31,096 small matrices and compare full success with both the new source peeling and the retained capped planner. All agree. A further 350 seeded random matrices are checked against every row-subset stopping condition and the permanent-one-minor characterization. These tests include rectangular inputs, zero rows, coefficients two, and incomplete prefixes. The explicit 4,000-row chain test records exactly 8,000 source-incidence visits and zero fill updates.

The seven new column-cancellation test methods are included in a focused 36-test run with the existing core, whole-cut, primitive-annulus and disk-absorption tests. All 36 pass. Certificate tests cover duplicate pivots, nonexistent and Boolean labels, malformed counts, altered certificates, large integers, prefix checking and caller aborts.

The saturated extension adds 17 focused tests, all passing. They include native geometric sources, independent replay with the producers disabled, altered port charts, exact binary/hexadecimal transport, budget exhaustion, and callback propagation. Negative controls include the aggregate-two annulus and the winding-three Möbius attachment described above. An abstract finite interface model takes base genus

$$q = 2^{16384} + 17$$

and obtains genus $2q$ using one aggregate incidence. This checks the implementation's refusal to expand numerical genus and its exact integer transport. It is a conditional finite-model test, not a

claim that a new native triangulation realizing that particular huge-genus source was constructed during this study.

The passenger extension has ten additional tests. An independent oracle performs the full positive update on all remaining rows, rather than using immutable donors. It examines all 729 ternary 2×3 donor matrices and 8,748 candidate complete schedules; the 324 valid schedules agree exactly. There are also 120 randomly relabelled acyclic examples, branching and merging paths, passenger-only columns, empty cases, and uncapped counts exceeding 8,192 bits. Invalid schedules are rejected.

The maintained suite run completed **1,687 tests in 365.654 seconds**, all passing. The passenger module was added after that run's collection and was tested separately: all ten tests passed. A final collection-only check counts 1,697 tests in the delivered source tree. Thus the reported validation is one complete 1,687-test run plus the ten new isolated tests, not a claimed second full 1,697-test execution. The maintained run also reports 13,846 expanded-sheet topology comparisons and its independent marked-cover audit; the full output is retained verbatim.

7.2 The pinned source corpus and unchanged core coverage

The retained corpus consists of 106 supplied normal cuts, including layered solid-torus cases, mixed coordinates, source relabellings, other finite triangulations and large binary coordinate families. Its exact JSON has SHA-256

`97b4ceed6f068885760d80dea214538634998dd405b77e76926c1f93db03da5c`.

Every experiment names the repository pin. The saved old primitive-annulus audit is also hashed. This corpus is a retained regression collection, not a random sample of knots and not a performance distribution for crossing number n.

On prepared source cores, the new producer follows these paths:

Core outcome or certificate path	Cases
Existing square row-leaf success	66
New column-leaf certificate path	24
Raw-cancellation stall	6
Core three-handle exclusion	10

The 24 column-path successes were already successes of the old capped planner. Both produce 90 successful core results. Substituting the new core producer into the existing whole-cut consumer gives exactly the same 78 complete censuses and 28 inconclusive results as before; all 156 successful old/new whole-cut certificates independently replay. This confirms the acceptance-class correction proved in Proposition 3.3.

7.3 Paired timing protocol and synthetic scaling

The benchmark retains exact copies of the old producer and verifier from the pinned commit. It alternates execution order over five paired trials, uses a preliminary measurement to choose a small repeat count for each side, divides by that repeat count, and reports the median per call. The ratio is the baseline median divided by the new median. Raw samples, repeat counts and operation counters are in the delivered JSON.

The recorded environment is CPython 3.12.14 on Linux 6.18.44, x86_64, with glibc 2.39. The experiments used the shared research runtime; they are not hardware-isolated microbenchmarks or

statistical confidence intervals. Their strongest asymptotic evidence is the source-operation count, alongside the proved bound.

For the bidiagonal family N_m of Section 3, the old planner records $2m(m+1)$ source-entry scans and the new planner $2m$ incidence visits. Both perform zero fill updates. The measured producer medians are:

Rows m	Nonzeros	Baseline (ms)	New (ms)	Ratio
16	32	0.1074	0.0190	5.67×
64	128	1.4318	0.0764	18.73×
256	512	20.1221	0.5037	39.95×
1,024	2,048	552.0291	2.3910	230.88×
4,096	8,192	7,456.9916	7.2578	1,027.45×

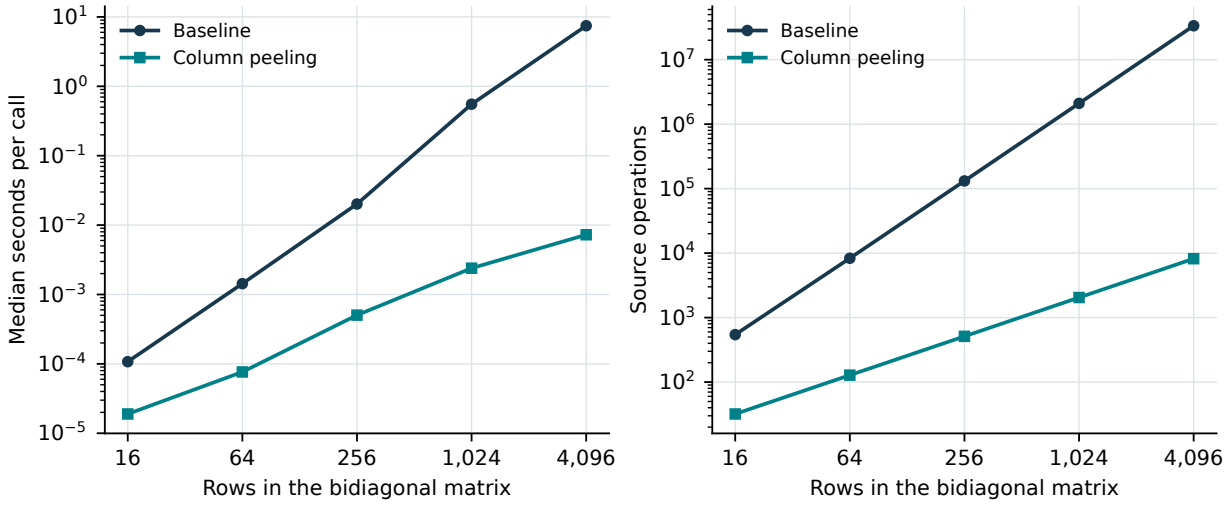


Figure 1: The measured bidiagonal cancellation family. Left: five-trial median producer times. Right: exact source-operation counters. Both axes are logarithmic. The curves connect measured points; they are not fitted claims about the full recognizer. The figure is regenerated directly from the supplied benchmark JSON.

The synthetic replay experiment uses conditional core incidence words, not authenticated embedded three-dimensional sources. At 256 rows its old median is 729.837 ms and its new median is 0.776 ms, a ratio of 939.98×. This shows the cost removed by immutable timestamp replay in the intended finite checker. It does not measure source authentication.

7.4 Native timing: a small-kernel regression and almost unchanged queries

The source-backed timing experiment separates prepared finite cores from the entire old whole-cut query:

Scope	Baseline	New	Ratio
All 106 prepared core producers	4.807 ms	6.515 ms	0.738×
The 24 affected core producers	1.397 ms	0.974 ms	1.435×
All 90 successful core replays	7.084 ms	5.703 ms	1.242×
Complete 106-case old cut query	36.515 s	36.211 s	1.008×

The aggregate prepared-core timing is a regression: the new median is about 35.6% larger. This result must remain visible alongside the affected subset's improvement. The samples vary, and such small calls are sensitive to runtime overhead. The retained fast row-leaf path helps preserve compatibility, but the data do not support a universal practical speedup.

The complete-query medians are effectively unchanged. Source orbit construction, boundary observation and other work remain substantial; the raw finite planner is too small a fraction of this corpus's full query cost to transform the total. The synthetic thousand-fold ratio must therefore not be advertised as an unknot-recognition speedup. A next performance study should target authenticated source-query reuse and boundary observation, while preserving this paired full-query scope.

7.5 Seven additional geometric successes

The saturated-bundle operator addresses coverage rather than simply substituting a faster planner. Its final independent audit has the following outcomes:

Audit quantity	Count
Supplied cuts	106
Complete new whole-cut censuses	**85**
Earlier complete censuses retained	78 of 78
New complete censuses	7
Independent source-certificate replays	85 of 85
Fresh Regina literal cuts completed	99
Successful source censuses within literal-cut cap	78
Agreement on those successful censuses	78 of 78
Large-coordinate successes replayed without expansion	7
Remaining joint raw stalls	11
Remaining three-handle exclusions	10

All seven new cases previously returned `annular_core_attachment`. All seven admit fresh literal cuts in Regina 7.4, and those independently computed censuses agree with the new source proof. The seven successful large-coordinate cases beyond the literal expansion cap are distinct from these seven new cases; they are independently replayed but are not claimed to have fresh expanded Regina comparisons.

In the following table b is the number of base boundary components. The listed genera are for the whole cut, with repeats indicating distinct components. Genus zero denotes a ball.

Case	T	Saturated nondisk base	Cut genera
`pilot/fibonacci_lst_05/0`	5	Möbius, $(\chi, b) = (0, 1)$	$1, 2$
`pilot/fibonacci_lst_07/0`	7	Möbius, $(0, 1)$	$1, 2$
`pilot/fibonacci_lst_09/0`	9	Möbius, $(0, 1)$	$1, 2$
`pilot/lst_4_7/0`	4	Möbius, $(0, 1)$	$1, 2$
`pilot/lst_7_10/0`	5	Möbius, $(0, 1)$	$1, 2$
`fibonacci_lst_09/mixed`	9	One crosscap, $(-1, 2)$	$0, 3, 3$
`fibonacci_lst_10/mixed`	10	Annulus, $(0, 2)$	$0, 2, 2$

The entire audit, including fresh oracle subprocesses, took 86.04 seconds. That is a single mixed validation run, not a paired speed comparison with the prior consumer. The package retains every case outcome, refusal reason, source hash and fresh-oracle result. All 21 unresolved cases remain inconclusive.

7.6 Explicit witnesses for the remaining raw stalls

The package also extracts all eleven remaining joint stopping sets. For each case, the parallelity and boundary source proofs independently replay before the exact matrix is read, and the separate obstruction checker verifies the resulting nonempty row set. Six cases already have an obstruction in their original-core row submatrix. In the other five, that submatrix fully cancels and the obstruction requires the added saturated relations. All five of those latter cases have saturated disk bases, so the unresolved compatibility is already present in the disk-absorption subproblem.

For example, `solid_torus_sum_rp3/mixed` has the stopping rows $(8, 10, 11)$ on its nonzero stopping columns $(8, 28, 32)$, with matrix

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Every column sum is two. Its original-core row submatrix nevertheless admits a full raw schedule. This small source witness makes the joint compatibility issue concrete. It certifies a raw obstruction; it does not infer the topology of that cut from an unsigned determinant or from the triangulation's example name.

Exact uncapped matrices, row ownership, canonical stopping sets, source-proof hashes and the old-core comparison are retained in `artifacts/saturated/stopping_obstructions.json`. A portable extractor reproduces them. These witnesses make the next investigation of refused cases reproducible, while keeping unsupported topological conclusions separate from algebraic ones.

7.7 A complete small source example

The smallest new example is `pilot/lst_4_7/0`. Its triangulation has isomorphism signature `eHKPbcddhqxr` and four tetrahedra. Its normal coordinates, in the source's seven-coordinate order per tetrahedron, are

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 \end{pmatrix}.$$

The new saturated base is profile 0, a Möbius band with one fully attached annulus. It contributes one virtual band and relation row 5. On original nonforest columns $(4, 12, 14, 15, 18, 19, 20, 21)$, the exact aggregate matrix, in row order $0, \ldots, 5$, is

$$A = \begin{pmatrix} 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The omitted bundle column contains two in row 5 and zero in every other row. The first five rows are the original core relations; they already admit full cancellation when considered alone. The verified source row/column schedule is

$$(2, 14),\ (5, 4),\ (3, 21),\ (0, 15),\ (4, 20),\ (1, 19).$$

The result consists of two components, of genera two and one. Their Euler characteristics are -1 and 0, summing to -1. Relation 5 is cancelled at the second step using an old core column, while its virtual band never appears in the stored matrix. This illustrates the exact mechanism of the geometric extension on a genuine input.

The complete labelled tetrahedron gluings, source proofs, forest, boundary charts, finite certificate and normal vector are in `artifacts/saturated/certificates/019.json`. They determine the source even if a different triangulation labelling is produced from the isomorphism signature. The other six cases are recorded at the same level of detail, rather than only by an expected genus or an example name.

8 What the reduction contributes to the hierarchy goal

8.1 A smaller exact local problem

The saturated reduction replaces a potentially huge explicit relative handle system by the matrix

$$A \in \mathbb{N}^{(f+s)\times c}, \qquad |\operatorname{supp} A| \le I_0 + P.$$

The new band handles affect genus counts and a geometric existence proof, but they do not become cancellation choices. Complete source cancellation is a deterministic sparse operation. A failure is accompanied by a coordinate-dependent stopping obstruction to the entire raw rule.

These are useful improvements for a hierarchy implementation because they give a precise accounting unit: the finite inherited core and active interfaces, with numerical topological data kept in binary. They also exclude an unproductive source of search. For a fixed matrix, branching over its positive raw pivot orders or proposed survivors cannot discover a full cancellation when the complete column test fails.

The result is not a bound on how many different matrices a hierarchy must visit. A new normal surface, a different disk system or a genuinely intersection-reducing move can change the stopping set. Controlling those changes remains a geometric question.

8.2 A precise global accounting criterion

It is helpful to state a global cost target in a form that does not confuse deterministic work with branching. The following elementary search-tree lemma is an accounting device, not a new unknot-recognition theorem.

Lemma 8.1 (Branch-product bound)**.** *Let a finite rooted search tree have maximum depth D. At an internal vertex v, let $b(v) \ge 1$ be its number of children. Suppose every root-to-leaf path satisfies*

$$\sum_{v \text{ on the path}} \log_2 b(v) \le K.$$

Then the tree has at most 2^K leaves and at most $(D+1)2^K$ vertices.

Proof. Start with mass one at the root and divide each vertex's mass equally among its children. A leaf receives $\prod_v b(v)^{-1} \ge 2^{-K}$, while the leaf masses sum to one. Thus there are at most 2^K leaves. Every vertex lies on some root-to-leaf path, and the union of those paths has at most $(D+1)$ vertices per leaf. □

Theorem 8.2 (A sufficient quasi-polynomial integration contract)**.** *Suppose a complete deterministic recognition procedure on an n-crossing diagram has a finite search tree with the following properties:*

1. *Every explored state, source certificate and transported boundary pattern has encoded size at most n^a for a fixed constant a.*
2. *Each node's full processing cost, including source construction, all rejected local candidates and independent local certification, is polynomial in n.*
3. *The depth is polynomial in n, and the sum of $\log_2 b(v)$ along every root-to-leaf path is $O((\log n)^2)$.*
4. *Every terminal conclusion is sound, and the search covers all required alternatives, so completed exhaustion justifies its negative conclusions.*

Then the procedure runs in $n^{O(\log n)}$ bit time. The cancellation operators developed in this report satisfy a polynomial local-cost requirement once their authenticated source data is available.

Proof. Lemma 8.1 bounds the node count by $\mathrm{poly}(n)2^{O((\log n)^2)} = n^{O(\log n)}$. Multiplying by the polynomial per-node bit cost preserves this bound. The last hypothesis supplies decision correctness. The finite cancellation bound follows from Theorems 3.1 and 5.2, Proposition 6.1, and polynomial binary arithmetic for the census. □

The hypotheses are intentionally explicit. They have *not* been proved for the maintained general recognizer. A polynomial number of deterministic steps along a branch would be harmless under this contract; an uncontrolled number of branching choices or a large rebuild cost would not be. Binary compression of one supplied cut addresses only part of the first two items.

Lackenby's hierarchy work provides the mathematical framework for further development[6, 4]. Efficient existence and verification statements must be distinguished from finding the desired surface and from bounding all unsuccessful work. A short certificate does not by itself give a comparably fast deterministic producer.

8.3 Remaining obligations with concrete failure modes

Surface coverage. A supplied-cut census says nothing about whether a producer can find a useful surface within a small collection of sectors. Dense normal vectors and large coordinates may be perfectly manageable once supplied; they can still be difficult to discover. A complete theorem must quantify the entire candidate set, including rejected candidates.

Marked transport. The new proof establishes an unmarked homeomorphism type and raw incidence properties. A later hierarchy step needs the inherited boundary pattern as embedded curves and graphs. Unsigned passenger transport supplies exact numerical data under its update rule, but does not retain cyclic order or identify which endpoints join. These missing data can affect essentiality of a terminal pattern.

Escaping a raw stopping set. Example 2.11 shows that even a ball can have a blocked chosen raw system. A complete algorithm therefore needs operations that can lower actual geometric intersections or change the chosen source decomposition. Pure positive pivots cannot do this. The required repair cost must be bounded on all instances, not inferred from a successful finite corpus.

Three-handles and closed bases. The present consumer excludes both. A source core with a three-handle does not become simpler merely because its boundary genera are known. A valid extension requires actual geometric removals or a separate certified topological argument.

State identity and restarts. The canonical stopping set belongs to one labelled source matrix. It does not provide canonical representatives of embedded handle structures. Memoization can be sound only if its keys retain the needed geometric and marking information. Rebuilding after a compression or changing the forest can introduce new states; their number and encoded sizes must be included in the global sum.

Diagram provenance and final decisions. The normal-cut source contracts bind a triangulation and normal vector. They do not independently bind an arbitrary diagram to that exterior. The exact full recognizer retains its existing routes and fallback. The experimental APIs in this package issue supplied-cut results, not new standalone knot verdicts.

8.4 A realistic next milestone

The most direct continuation is a *marked* version of the complete column certificate. Its no-fill order has immutable donors and a directed acyclic dependency graph. This removes uncertainty about pivot discovery and limits numerical growth, allowing the geometric work to focus on transporting embedded strand data through a fixed successful schedule.

A useful milestone would accept an authenticated source core and a finite marked pattern, produce the entire surviving marked handlebody structure, and independently replay every strand attachment without expanding large parallel families. Combining that with one fully certified next hierarchy cut would demonstrate a genuine transition, rather than another isolated topological census. It would still need a global coverage and branching argument to achieve the quasi-polynomial target.

9 Further research questions and proposed next work

The following questions are ordered by how directly they build on the proved and implemented results. Each asks for a concrete mathematical or computational advance, with a criterion by which progress can be assessed.

9.1 Transport an embedded boundary pattern through a no-fill schedule

Can the immutable donor structure be lifted from unsigned rows to ordered strand systems on the boundary of the handlebody? The input should specify actual source endpoints, local cyclic orders and fibre transport, not only intersection counts. The output should encode the induced pattern after each geometric cancellation or directly on the final handlebody.

The passenger theorem bounds the growth of numerical counts. The unresolved issue is the geometry of their concatenation. A successful result should give a polynomial bound on the size of a shared representation and an independent replay theorem. Small counterexamples with equal incidence and different boundary order should be part of the validation set.

9.2 Give saturated bands a compressed marked realization

The saturated theorem omits a potentially enormous family of even band columns because they cannot be pivots. Can one retain those bands as a small typed family while transporting specified markings through them? The target is a source-certified relative band system, with attachment addresses and cyclic order, whose encoding is polynomial in the source and the bit length of the base topology.

An abstract free-group presentation with a large product of commutators does not settle this geometric problem. The representation should support both orientable and nonorientable bases and arbitrary compatible annular gluing twists. The punctured-torus and Möbius examples provide elementary test cases, while the seven new source cuts provide native regression cases.

9.3 Identify small repairs of a raw stopping core

Which geometric moves destroy a specified stopping set by decreasing actual intersections? The persistence theorem proves that searching more positive pivot orders cannot help. A useful repair theorem would identify a bounded-size configuration in every blocked core belonging to a specified topological class and supply a verified transformation that reduces a stated potential.

The class should be restricted first. For example, study layered solid tori or source cuts with a bounded number of saturated relations. The goal is a theorem with explicit dependence on the chosen parameter, not a promise inferred from random small examples. The blocked ball matrix $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ is the first required test: a useful repair can undo a handle slide, but no positive raw pivot order can finish it.

9.4 Optimize the choice of source forest with certified scope

Our completeness theorem fixes the core forest. How does $\mathrm{Stop}(A)$ change when a tree edge is exchanged with a nonforest edge? An incremental exact incidence update could avoid reconstructing the entire source for every forest. The difficult part is to prove the geometric cocore correspondence and to bound a complete family of forests or exchanges.

A practical first deliverable is a source-certified one-edge exchange with an independently verified new aggregate matrix. A later theorem could characterize a class in which polynomially many exchanges expose full raw cancellation. The mere existence of a successful forest does not give a fast complete search over all forests.

9.5 Use a stopping certificate to focus higher-level moves

The residual is a canonical row set in a fixed matrix. Can repair proposals be restricted to neighborhoods of that set while preserving completeness? One possibility is to keep a typed incidence graph containing its rows, their columns and the nearby geometric charts. A theorem would need to show that irrelevant regions can be retained by identity while a useful repair is found inside the selected region.

This is a locality question about geometric simplification, not a consequence of the algebraic stopping condition alone. Its benefit would be a smaller complete move neighborhood with an explicit certificate of unchanged exterior data.

9.6 Extend the consumer to three-handle cores

Ten cuts in the retained audit still have core three-handles. Can a dual source peeling rule identify geometric two/three-handle cancellations without flattening the compressed cut? A suitable rule

must authenticate the attaching sphere and its unique intersection, rather than treating an algebraic incidence as sufficient.

Begin with cases where a genuine free face is present and record its effect on all interface charts. Then investigate whether a bounded sequence of dual moves covers a useful larger class. Every accepted reduction should have an independent source replay and preserve the component census used by the present implementation.

9.7 Resolve the eleven remaining raw stalls

The source audit gives eleven cuts with a failed combined raw system. Their exact canonical stopping witnesses are included: six obstructions already lie in the original core, and five require added saturated disk relations. Classify the geometry behind those witnesses: shared annular powers, interacting original two-handles, graph cycles, or another configuration. This is a finite, concrete next experiment.

The desired output is not merely a longer list of heuristic moves. It is a small set of proved geometric lemmas, each supported by source certificates, that either resolves a family of stalls or explains why a new surface or disk system is needed. Some test cuts may actually fail to be handlebodies; in those cases a separate authenticated topological obstruction is the appropriate goal. Any cases still unsupported should remain inconclusive.

9.8 Unify geometric source compilation with the incoming interface kernels

The moment-coded first-return report compresses boundary observations, while the present operator consumes their ordered annular words. Can the source proofs be reused without changing their trust boundary, reducing the repeated orbit-query cost that dominates small examples?

The interface contract must preserve shared corners, actual gaps, orientation transport, whole-circle detection and port ownership. A useful implementation should retain the old query as a differential oracle and record both complete-query and finite-kernel timings. This question is promising precisely because the current empirical evidence does not show a broad whole-query speedup from the new core planner alone.

9.9 Find a useful-surface coverage theorem with an explicit budget

Can every nonterminal member of a restricted hierarchy class be advanced by a surface whose active interface, saturated residue and repair cost have a provable bound? Candidate parameters include the number of saturated relations and the geometry of their stopping core. A useful theorem should specify how the surface is found, how failures are charged and how the inherited pattern is transported.

An existence theorem without a producer or an upper bound on failed candidates is insufficient for the running-time target. Conversely, a complete producer on a clearly defined restricted class would already be a substantial result, even before extension to all knot exteriors.

9.10 Prove a global branching and state-size potential

Find a potential controlling branching, resets and encoded state size. Theorem 8.2 asks for an $O((\log n)^2)$ path branching budget with polynomial depth and per-state work.

The potential must include boundary patterns, geometric repairs and source rebuilding. A raw stopping core alone cannot suffice: coordinate changes can create or remove it. Proving this combined bound is the principal remaining step toward a general $n^{O(\log n)}$ recognizer.

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A Package contents, integration, and reproduction

A.1 Layout and source identity

The archive is rooted at `ProveIt_Unknot_Stopping_Cores_2026-10-12/`. Its top-level README describes the recommended integration order and the scope of each API. The important directories are:

`article/`
This article's complete TeX sources, reference list, compiled PDF, vector and raster figure, figure-generation script and build script.

`integration/`
A repository-relative patch, complete updated source files, old/new file hashes and checked integration instructions. Existing code changes are limited to the core producer and verifier; the saturated and passenger APIs are added modules.

`artifacts/column_cancellation/`
The complete paired benchmark JSON and focused test log. The reproducible benchmark driver and exact baseline modules are part of the repository integration overlay.

`artifacts/saturated/`
The full 106-case audit, environment, source provenance, exact seven new cases and their source certificates, the worked matrix for case 019, focused test log and portable audit driver.

`artifacts/validation/`
The complete maintained-suite log, separate passenger test log and validation summary.

`artifacts/source/`
The exact retained 106-case corpus, previous primitive-annulus audit, and their provenance hashes.

The source integration targets repository revision

`704600626606a9097cfd89d4e3d2bbc63c8dd1cb`.

A later revision requires reviewing the affected interfaces and replaying the retained evidence. The package does not include a duplicate repository checkout. Its patch was checked against exact pinned base files in an isolated tree. The complete replacements provide a convenient second way to inspect every proposed source change.

A.2 Apply and validate the source changes

Let `PROVEIT_CHECKOUT` be the root of a checkout at the pinned revision, and `RESEARCH_PACKAGE` the root of this unpacked package. Review `integration/README.md` and its old/new hash manifest first. The essential Git operations, run from the checkout root, are:

```
git apply --check "$RESEARCH_PACKAGE/integration/integration.patch"
git apply "$RESEARCH_PACKAGE/integration/integration.patch"
```

These commands are provided for the maintainer's integration workflow. This research run made no remote commit or repository publication.

Run the tests from `Topology/UnknotRecognition/fast` in the patched checkout:

```
python3 -m unittest tests.test_column_cancellation \
    tests.test_normal_cut_saturated \
    tests.test_raw_passenger_transport -v
python3 -m unittest discover -s tests
```

The finite producer and replay code use Python's standard library and the existing ProveIt modules. The maintained repository's broader tests and oracle comparisons retain their existing dependency requirements. The retained fresh literal-cut audit used Regina 7.4.

A.3 Reproduce the paired benchmark

From the same `fast` directory:

```
python3 benchmark_column_cancellation.py \
    --trials 5 --output reproduced-column-cancellation.json
```

The driver locates its baseline modules and corpus relative to its own file. It compares the same source cores and verifies the successful results before recording timings. The optional `-skip-whole-cuts` flag is useful for a finite-kernel-only experiment, but such a run cannot reproduce the complete-query row in the article's table. The largest synthetic baseline and the complete query make the full experiment appreciably longer than a unit test.

Timing samples are environment-dependent. Reproduction should confirm the algebraic outcomes, source censuses and operation counters; it need not reproduce the saved wall-clock numbers digit for digit. A proposed optimization should retain both affected-subset and all-case timings, including regressions.

A.4 Reproduce the saturated source audit

The packaged driver accepts an explicit checkout path:

```
python3 "$RESEARCH_PACKAGE/artifacts/saturated/run_audit.py" \
    --checkout "$PROVEIT_CHECKOUT" --fresh-oracles \
    --output reproduced-saturated/audit.json
```

It reruns the source producer, disables producer entry points during independent replay, preserves earlier successes, and invokes the maintained Regina literal-cut oracle subject to its expansion cap. Omitting `-fresh-oracles` uses the historical comparison packets while still rerunning the new source pipeline. The output directory is separate from the retained audit by default.

For a single retained certificate, use the source wrapper's verifier:

```
import json
from fastunknot.normal_cut_saturated_verify import \
    verify_normal_cut_saturated

with open("/path/to/certificates/019.json") as source:
    case = json.load(source)
assert verify_normal_cut_saturated(
    case["triangulation"], case["coordinates"],
    case["certificate"])
```

The certificate contains the labelled source gluings. No inference from the isomorphism signature or from the expected genus is needed.

A.5 Call the new operators

The saturated wrapper is opt-in:

```
from fastunknot.normal_cut_saturated import normal_cut_saturated
from fastunknot.normal_cut_saturated_verify import \
    verify_normal_cut_saturated

answer = normal_cut_saturated(
    triangulation, coordinates, record_certificate=True)
if answer["status"] == "COMPLETE":
    assert verify_normal_cut_saturated(
        triangulation, coordinates, answer["certificate"])
    histogram = answer["census"]["histogram"]
```

The lower-level `saturated_source_census` is conditional on already authenticated geometry, base summaries and an ordered boundary skeleton. The wrapper supplies that authentication.

For exact count transport after a complete column schedule:

```
from fastunknot.core_handlebody import column_cancellation_plan
from fastunknot.raw_passenger_transport import transport_raw_passenger

counts = {0: {0: 1, 1: 2, 2: 3},
          1: {1: 1, 3: 5}, 2: {2: 1, 3: 7}}
plan = column_cancellation_plan(counts)
passenger = transport_raw_passenger(
    counts, plan["cancellations"], {0: 11, 3: 13})
assert passenger == {3: 354}
```

The original exact `counts` must be supplied, not a capped copy. The operation verifies its donor schedule but does not authenticate embedded passenger geometry.

A.6 Build the article and figure

From any working directory, run

```
python3 /path/to/package/article/build.py
```

The script uses `latexmk` when available and otherwise runs `pdflatex` three times. It needs a standard TeX installation with the packages named in the source. All included figure files are already built. To regenerate them from the retained benchmark JSON, run `article/make_figures.py` with Matplotlib installed, then rebuild the article. The delivered PDF was compiled, checked for unresolved references and layout defects, rendered, and visually inspected.

A.7 Interpretation and integration limits

The patch upgrades raw planning and replay, and adds a broader positive source-cut census. It does not change the recognizer's dispatcher or its exact fallback policy. A caller must preserve `INCONCLUSIVE` as an unsupported or unfinished result. A stopping-set certificate excludes only a complete raw schedule in its named source matrix.

The next integration milestone is to preserve embedded boundary data through these complete schedules and certify a subsequent hierarchy transition. The present article supplies precise local theorems, implementations and evidence on which that work can build; the global quasi-polynomial claim remains a research objective.